Achieving Higher Factual Accuracy in Llama LLM with Weighted Distribution of Retrieval-Augmented Generation

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Abstract

Introducing a novel concept, the integration of a weighted distribution of Retrieval-Augmented Generation (RAG) with the Llama Large language model significantly enhances factual accuracy and contextual relevance in generated text. Experimental results show substantial improvements in precision, recall, F1 score, and BLEU score, demonstrating the effectiveness of the weighted RAG mechanism in prioritizing high-quality information during the generation process. Human evaluations further validate the model's practical applicability and reliability, highlighting its potential for deployment in high-stakes environments. The contributions of this research provide a scalable framework for improving language models, offering new avenues for dynamic context-aware weighting and real-time feedback integration. Future work will focus on refining the weighting mechanism, exploring advanced retrieval algorithms, and expanding applications to multilingual settings and domain-specific corpora, driving continued innovation in natural language processing.

Keywords: Retrieval-Augmented Generation, Llama Large, Factual Accuracy, Contextual Relevance, NLP, Language Models

1. Introduction

The Llama Large Language Model (LLM) represents a significant advancement in the field of natural language processing, leveraging extensive training on diverse datasets to achieve impressive language understanding and generation capabilities [1, 2]. Despite its robust architecture and high performance on a variety of tasks, ensuring factual accuracy remains a critical challenge [1, 3]. The ability of large language models to generate plausible but incorrect information poses substantial risks, particularly in applications requiring precise and reliable outputs [4]. Retrieval-Augmented Generation (RAG) emerges as a promising technique to address this challenge by integrating retrieval mechanisms into the generation process. RAG enhances language models by augmenting them with relevant information retrieved from external databases, thereby grounding their outputs in factual data [5]. This hybrid approach not only improves the relevance and accuracy of generated text but also expands the model's effective knowledge base beyond its initial training data.

The motivation for incorporating a weighted distribution of RAG lies in its potential to fine-tune the balance between generative creativity and factual correctness. By assigning weights to different retrieved documents based on their relevance and reliability, the system can prioritize high-quality sources, leading to more accurate and trustworthy responses. This approach aims to optimize the performance of Llama Large by systematically refining its output generation process, thus mitigating the issue of misinformation.

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The Google BIG-Bench dataset provides a comprehensive benchmark for evaluating the effectiveness of this enhanced model. BIG-Bench encompasses a wide array of tasks designed to test the capabilities of language models across various domains, including factual accuracy, reasoning, and comprehension. Utilizing this dataset enables a rigorous assessment of the improvements brought about by the weighted RAG integration, offering insights into its practical benefits and limitations.

The major contributions of this article are:

1. Introducing a novel weighted distribution mechanism in RAG to enhance the factual accuracy of language models.
2. Demonstrating substantial improvements in precision, recall, F1 score, and BLEU score through comprehensive experiments.
3. Providing a scalable framework that can be adapted to various language models and applications.
4. Highlighting the practical applicability of the enhanced model through human evaluations.
5. Addressing the inherent limitations of traditional LLMs by combining retrieval and generation paradigms.
6. Offering insights into future research directions, including dynamic context-aware weighting and real-time feedback integration.

This paper is structured as follows: The introduction sets the stage by outlining the challenges and motivations behind the research. The related work section reviews existing methodologies and contextualizes our approach within the broader landscape of language model enhancement. The methodology section details the architectural modifications and experimental setup, including a thorough explanation of the weighted RAG mechanism. Results and analysis present empirical findings, demonstrating the performance gains achieved through our approach.

The discussion interprets these results, highlighting their implications for the field and suggesting potential directions for future research. The conclusion summarizes the contributions and key takeaways, while the appendix provides supplementary material and additional experimental data.

2. Literature Survey

2.1. LLM Hallucination

LLMs frequently generated outputs that were fluent and coherent yet factually incorrect or nonsensical, a phenomenon termed hallucination [6]. Hallucinations in LLMs emerged due to the models' reliance on probabilistic predictions derived from extensive training data, which often led to the creation of plausible but false information, posing significant risks in applications requiring high reliability [7]. Mitigating hallucination involved several strategies, including fine-tuning on domain-specific data, which aimed to improve the factual accuracy of model outputs by narrowing the model's focus [7]. Another approach leveraged external knowledge bases, integrating structured data into the generation process to enhance the grounding of outputs in verifiable information [8]. Post-generation verification methods, such as fact-checking, were also employed to identify and correct inaccuracies in real-time, thereby enhancing the trustworthiness of the models [9]. Evaluating the effectiveness of these mitigation strategies posed its challenges, as traditional metrics like precision, recall, and F1 score were supplemented with human evaluations to better capture the nuances of factual accuracy in generated content [10]. Newer metrics, specifically designed to quantify hallucination rates, provided a more nuanced understanding of LLM performance, aiding in the development of more reliable language models [7]. The collective goal of those efforts was to enhance the factual reliability of LLMs, reducing the incidence of hallucinations and improving their applicability in critical domains.

2.2. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is as a powerful technique for enhancing the factual accuracy of language models by combining retrieval-based and generative approaches [11, 12]. RAG operated by retrieving relevant documents from a large corpus and incorporating this information into the generation process, thereby grounding the output in external knowledge and improving its accuracy and relevance [13]. Various implementations of RAG were proposed, each differing in how retrieval and generation were integrated. Some models employed a simple retrieval mechanism that selected top-k documents based on relevance scores and used them directly in the generation step [11, 14, 15]. Others implemented more sophisticated weighting mechanisms that prioritized high-quality sources, ensuring that the most reliable information influenced the generated text [16]. The choice of retrieval corpus and the method of integrating retrieved information significantly affected the performance of RAG systems, highlighting the importance of careful design in these areas [17]. Challenges in RAG included ensuring the relevance and reliability of retrieved documents, efficiently handling large-scale retrieval, and seamlessly integrating this information without compromising fluency [18, 19]. Research efforts focused on improving retrieval algorithms, enhancing the interpretability of retrieved information, and developing hybrid models that dynamically balanced generative creativity and factual accuracy [20, 21]. Those advancements aimed to leverage the strengths of both retrieval and generation, resulting in language models that produced more accurate and contextually relevant outputs.

2.3. LLM Benchmarks

Benchmarks played a crucial role in evaluating the performance of LLMs across a variety of tasks, providing standardized datasets and evaluation protocols that enabled fair comparisons between different models and approaches [22]. The Google BIG-Bench, in particular, offered a comprehensive assessment of LLM capabilities, covering a wide range of domains and task types, including factual accuracy, reasoning, and comprehension [23]. Benchmark datasets like BIG-Bench included tasks specifically designed to test the limits of language models, providing a rigorous testing ground for assessing improvements and identifying areas for further development [23, 24]. Other popular benchmarks, such as the GLUE (General Language Understanding Evaluation) and SuperGLUE datasets, focused on natural language understanding tasks, offering additional insights into the performance of LLMs on specific challenges [25]. Evaluation metrics commonly used in benchmarks included accuracy, F1 score, and BLEU score, along with human evaluations for tasks requiring subjective judgments [26]. The metrics ensured that models were tested under consistent conditions, facilitating meaningful comparisons and driving advancements in the field [27]. The standardized evaluation protocols employed by benchmarks were essential for tracking progress and guiding future research, enabling the development of more capable and reliable language models [4, 28].

3. Methodology

3.1. Llama Large Language Model

The Llama Large Language Model (LLM) is built upon an extensive transformer-based architecture, which leverages multiple layers of attention mechanisms to capture complex dependencies within text data. Each layer in the transformer architecture contributes to the model's ability to understand and generate human-like language by focusing on different aspects of the input text. The model's training involves vast datasets that encompass a wide array of topics and linguistic styles, enabling it to generate coherent and contextually relevant responses. However, the Llama Large LLM also encounters limitations, particularly in maintaining factual accuracy. The model's reliance on probabilistic predictions from its training data often results in the generation of plausible but incorrect information. While the extensive training data improves the model's fluency and versatility, it does not inherently guarantee factual correctness. Addressing this limitation requires augmenting the model with mechanisms that ground its outputs in verifiable knowledge,

Algorithm 1 Retrieval-Augmented Generation (RAG)

Require: Query q, Corpus C
Ensure: Generated Output o

```
Retrieval Step
Compute similarity scores S(q, d_i) for each document d_i ∈ C
Select top-k documents D = {d_1, d_2, ..., d_k} based on S(q, d_i)
Generation Step
Initialize hidden state h_0 with q
for t = 1 to T do
    Compute context vector c_t using attention mechanism:
    c_t = Σ_{i=1}^{k} α_{t,i} d_i
    α_{t,i} = exp(e_{t,i}) / Σ_{j=1}^{k} exp(e_{t,j}), where e_{t,i} = h_{t-1} · d_i
    Update hidden state h_t = f(h_{t-1}, c_t)
    Generate output token o_t based on h_t
end for
Output o = {o_1, o_2, ..., o_T}
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thus enhancing its reliability in tasks demanding high factual accuracy.

3.2. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is a hybrid approach that integrates the advantages of retrieval-based and generative models to enhance the factual accuracy of language model outputs. RAG functions by initially retrieving relevant documents $D = \{d_1, d_2, \ldots, d_k\}$ from a large corpus C based on the input query q, and subsequently utilizing this retrieved information to guide the generation process G. The retrieved documents serve as a contextual and factual foundation, which assists the model in generating more accurate and contextually relevant responses. Previous implementations of RAG have demonstrated its efficacy in various applications, including question answering and dialogue systems, where grounding responses in external knowledge is critical. These implementations typically employ a two-step process: a retrieval step R that identifies the most relevant documents using similarity metrics S, and a generation step G that incorporates the retrieved information into the output. The results from these implementations have shown significant improvements in both the relevance and factual accuracy of the generated content, underscoring the potential of RAG to enhance the performance of large language models.

By employing this algorithm, RAG effectively grounds the generated content in relevant and reliable documents, which significantly enhances the factual accuracy and contextual relevance of the language model outputs. The retrieval step ensures that the most pertinent documents are selected based on similarity scores, while the generation step integrates this information through an attention mechanism, dynamically adjusting the influence of each document on the final output. This sophisticated integration of retrieval and generation processes allows RAG to leverage external knowledge effectively, resulting in outputs that are both accurate and contextually appropriate.

3.3. Weighted Distribution of RAG

The concept of weighted distribution in RAG involves assigning different weights w_i to the retrieved documents d_i based on their relevance and reliability. The theoretical framework for this approach is grounded in the idea that not all retrieved documents contribute equally to the generation process. By assigning higher weights to more relevant and reliable sources, the model can prioritize high-quality information, leading to more accurate and trustworthy outputs. Mathematically, this can be formulated by incorporating a weighting factor into the retrieval and generation process, which adjusts the influence of each retrieved document based on its assigned weight.

Given a query q and a set of retrieved documents $D = \{d_1, d_2, \ldots, d_k\}$, the weight w_i for each document d_i is determined by its relevance score r_i and reliability score ρ_i. The combined weight can be expressed as:

$$w_i = \frac{r_i \cdot \rho_i}{\sum_{j=1}^{k} r_j \cdot \rho_j}$$

where r_i represents the similarity score between the query q and the document d_i, and ρ_i denotes the reliability score of the document d_i.

The influence of each document on the generation process is then incorporated through an attention mechanism, where the context vector c_t at time step t is computed as:

$$c_t = \sum_{i=1}^{k} \alpha_{t,i} d_i, \quad \alpha_{t,i} = \frac{\exp(e_{t,i} \cdot w_i)}{\sum_{j=1}^{k} \exp(e_{t,j} \cdot w_j)}$$

Here, $e_{t,i} = h_{t-1} \cdot d_i$ represents the attention score, and h_{t-1} is the hidden state from the previous time step.

The implementation of weighted RAG with the Llama Large model involves several steps: first, a set of candidate documents is retrieved based on the input query; next, each document is assigned a weight based on its relevance and reliability; finally, the weighted documents are used to guide the generation process, with higher-weighted documents exerting a greater influence on the final output. The weighted influence of each document adjusts dynamically during generation, optimizing the balance between generative creativity and factual correctness, thus enhancing the overall performance of the Llama Large model in producing accurate and reliable responses.

This approach can be mathematically formulated as a constrained optimization problem, aiming to maximize the expected factual accuracy of the generated output o:

$$\max_{o} \mathbb{E}[\text{Accuracy}(o)] \quad \text{subject to} \quad \sum_{i=1}^{k} w_i = 1 \quad \text{and} \quad w_i \geq 0$$

where the constraints ensure that the weights are normalized and non-negative. By integrating the weighted RAG mechanism into the Llama Large model, the system effectively leverages the most relevant and reliable information, significantly improving the factual accuracy and contextual relevance of the generated outputs.

3.4. Experimental Setup

The experimental setup for integrating and testing the weighted RAG with the Llama Large model involves a meticulously designed environment to ensure rigorous evaluation. The setup includes various components, described as follows:

1. Hardware Specifications: High-performance GPUs and ample memory are essential to handle the computational demands of training and inference with large language models. The hardware setup includes multiple NVIDIA A100 GPUs with 80GB of memory each, interconnected through a high-speed NVLink interface, to facilitate efficient parallel processing and model training.
2. Software Environment: The software environment consists of advanced deep learning frameworks such as TensorFlow and PyTorch, which provide the necessary tools and libraries for model development and training. Additionally, specialized libraries for retrieval and generation tasks, such as Hugging Face's Transformers and FAISS (Facebook AI Similarity Search), are utilized to enhance the model's capabilities.
3. Dataset Preparation: The Google BIG-Bench dataset is curated to provide a comprehensive set of tasks designed to evaluate various aspects of language model performance, including factual accuracy, reasoning, and comprehension. The dataset is preprocessed to ensure compatibility with the Llama Large model and to facilitate efficient training and evaluation.
4. Training Protocol: The training protocol involves fine-tuning the Llama Large model with the weighted RAG mechanism. A combination of supervised learning and reinforcement learning techniques is employed to optimize the model's performance. Supervised learning involves using labeled data to guide the model towards generating accurate outputs, while reinforcement learning techniques are used to fine-tune the model's responses based on feedback from evaluation metrics.
5. Evaluation Protocols: Evaluation protocols include a series of benchmarks and metrics to assess the factual accuracy, relevance, and overall quality of the generated outputs. Metrics such as precision, recall, F1 score, and BLEU score are employed, alongside human evaluations, to provide a comprehensive assessment of the model's performance. The evaluation process involves systematically testing the model on the BIG-Bench tasks to demonstrate the efficacy of the weighted RAG approach.

By systematically testing the model on the BIG-Bench tasks, the experimental setup aims to demonstrate the efficacy of the weighted RAG approach in enhancing the Llama Large model's performance. This setup provides valuable insights into its practical applications and potential for further development, showcasing the improvements in factual accuracy and contextual relevance achieved through the integration of the weighted RAG mechanism.

4. Results and Analysis

4.1. Evaluation Metrics

The evaluation of the Llama Large model with the integrated weighted RAG mechanism employed a comprehensive set of metrics designed to measure factual accuracy and overall performance. Precision, recall, and F1 score were used to assess the accuracy of the model's outputs. Precision measures the proportion of relevant instances among the retrieved instances, recall quantifies the proportion of relevant instances that were successfully retrieved, and F1 score represents the harmonic mean of precision and recall. Additionally, BLEU (Bilingual Evaluation Understudy) score was used to evaluate the fluency and relevance of the generated text by comparing it to a set of reference texts. Human evaluations were also conducted to assess the contextual appropriateness and factual correctness of the model's outputs, providing a qualitative measure of performance that complemented the quantitative metrics.

These metrics collectively provided a robust framework for evaluating the improvements in the model's performance brought about by the weighted RAG mechanism. The combination of quantitative metrics and human evaluations offered a comprehensive assessment of the model, ensuring that both the factual accuracy and contextual relevance of the outputs were thoroughly examined.

4.2. Experimental Results

The experimental results demonstrated significant improvements in the performance of the Llama Large model with the weighted RAG mechanism. The model achieved notable increases in precision, recall, and F1 score across various tasks in the Google BIG-Bench dataset. The detailed performance metrics are presented in Table 2.

The weighted RAG model achieved an average precision of 89.3%, a recall of 85.7%, and an F1 score of 87.5%, significantly outperforming the baseline Llama Large model, which achieved an average precision of 82.6%, a recall of 78.9%, and an F1 score of 80.7%. The BLEU score for the weighted RAG model was 42.5, compared to 36.8 for the baseline model, indicating improved fluency and relevance of the generated text. Human evaluations further confirmed these findings, with the weighted RAG model receiving an average rating of 4.3 out of 5 for factual correctness and contextual relevance, compared to 3.7 for the baseline model. These results underscored the efficacy of the weighted RAG mechanism in enhancing the factual accuracy and overall performance of the Llama Large model.

4.3. Error Analysis

An analysis of the errors made by the model revealed several common patterns and limitations. The detailed error metrics are presented in Table 3.

One of the predominant errors involved the misinterpretation of ambiguous queries, where the model generated responses that were contextually relevant but factually incorrect. The weighted RAG model reduced such errors to 9.7%, compared to 14.2% for the baseline model. Another common error

Table 1: Evaluation Metrics for Llama Large Model with Weighted RAG

Metric	Description	Purpose
Precision	Proportion of relevant instances among the retrieved instances	Measures accuracy
Recall	Proportion of relevant instances successfully retrieved	Measures completeness
F1 Score	Harmonic mean of precision and recall	Balances precision and recall
BLEU Score	Evaluates fluency and relevance by comparing to reference texts	Measures text quality
Human Evaluation	Assesses contextual appropriateness and factual correctness	Provides qualitative assessment

Table 2: Performance Metrics Comparison between Baseline and Weighted RAG Models

Metric	Baseline	Weighted Model
Precision (%)	82.6	89.3
Recall (%)	78.9	85.7
F1 Score (%)	80.7	87.5
BLEU Score	36.8	42.5
Human Evaluation	3.7/5	4.3/5

was the over-reliance on certain high-weight documents, leading to biased outputs that did not fully consider the broader context. The weighted RAG model exhibited an 8.5% occurrence of this error, as opposed to 11.8% for the baseline. Additionally, the model occasionally struggled with highly specialized knowledge, producing outputs that were superficially accurate but lacked depth. The incidence of errors in specialized knowledge was 12.1% for the weighted RAG model, compared to 15.5% for the baseline. The limitations of the current approach were also evident in the handling of long-tail queries, where the model's performance was less consistent. The weighted RAG model handled long-tail queries with a 13.4% error rate, whereas the baseline model had an 18.7% error rate. Contextual irrelevance errors, where the generated output was not contextually appropriate, were reduced to 7.9% with the weighted RAG model, compared to 10.4% for the baseline. Superficial accuracy, where the responses appeared correct on the surface but were inaccurate upon closer examination, was observed at 9.3% for the weighted RAG model and 12.6% for the baseline. These findings highlighted the need for further refinement of the weighting mechanism and the incorporation of more sophisticated context-awareness strategies to mitigate these issues. Addressing these limitations is crucial for enhancing the robustness and reliability of the model in diverse applications.

5. Discussion

The experimental results indicate that integrating a weighted distribution of Retrieval-Augmented Generation (RAG) with the Llama Large model significantly enhances its performance, particularly in terms of factual accuracy and contextual relevance. The improved precision, recall, and F1 scores suggest that the weighted RAG mechanism effectively prioritizes high-quality information during the generation process. By assigning higher weights to more relevant and reliable sources, the model produces outputs that are not only accurate but also contextually appropriate, thereby addressing one of the major limitations of traditional LLMs. The observed increase in BLEU scores further corroborates the enhanced fluency and coherence of the generated text, underscoring the efficacy of the weighted RAG approach in refining the model's language generation capabilities.

The implications of these findings for the field of natural language processing (NLP) are profound. Enhancing the factual accuracy of LLMs addresses a critical challenge in their deployment for real-world applications, where the reliability of generated information is paramount. The weighted RAG mechanism offers a scalable solution that can be integrated into existing language models, thereby broadening the scope of their applicability. This approach can be particularly beneficial in domains that require high precision and reliability, such as legal document generation, medical information synthesis, and automated news reporting. By improving the factual grounding of LLM outputs, the weighted RAG mechanism contributes to increasing user trust and the overall utility of language models in professional and high-stakes environments.

The potential impact on future research and applications is considerable. Researchers can build on the weighted RAG framework to develop more sophisticated models that dynamically adjust the weighting of retrieved documents based on real-time feedback and evolving contextual cues. Future studies might explore the integration of more advanced retrieval algorithms and context-aware weighting mechanisms to further enhance the robustness and adaptability of language models. Additionally, the application of weighted RAG in multilingual settings could open new avenues for cross-lingual information retrieval and generation, thereby expanding the global reach and inclusivity of LLMs. The continuous improvement of factual accuracy in LLMs will likely drive innovation in various sectors, promoting the development of more reliable and contextually aware AI systems.

The observed error patterns highlight areas for further refinement. Misinterpretation of ambiguous queries and over-reliance on high-weight documents suggest that the weighting mechanism, while effective, may benefit from additional layers of context-awareness and disambiguation strategies. Incorporating semantic understanding and contextual reasoning capabilities could help mitigate these errors, leading to more nuanced and accurate responses. The challenge of handling long-tail queries and specialized knowledge underscores the need for domain-specific training and fine-tuning, which could be achieved through targeted data augmentation and specialized retrieval corpora. By addressing these limitations, researchers can further enhance the versatility and reliability of the weighted

Table 3: Error Analysis Comparison between Baseline and Weighted RAG Models

Error Type	**Baseline Model (%)**	**Weighted RAG Model (%)**
Misinterpretation of Ambiguous Queries	14.2	9.7
Over-reliance on High-Weight Documents	11.8	8.5
Errors in Specialized Knowledge	15.5	12.1
Long-tail Query Handling	18.7	13.4
Contextual Irrelevance	10.4	7.9
Superficial Accuracy	12.6	9.3

RAG model in diverse applications.

The practical applications of the weighted RAG mechanism extend beyond traditional NLP tasks. In educational settings, for instance, enhanced factual accuracy can improve the quality of automated tutoring systems, providing students with more reliable and contextually relevant information. In the field of customer support, more accurate language models can lead to improved automated responses, reducing the need for human intervention and increasing overall efficiency. Furthermore, in content creation and summarization, the ability to generate accurate and contextually rich outputs can significantly enhance the quality and reliability of automated content, benefiting industries such as publishing, marketing, and entertainment.

The integration of human evaluations alongside quantitative metrics in the experimental setup underscores the importance of a holistic approach to model evaluation. Human assessments of factual correctness and contextual relevance provide valuable insights that complement numerical scores, offering a more comprehensive understanding of model performance. This dual evaluation approach ensures that the improvements achieved through the weighted RAG mechanism are not only statistically significant but also practically meaningful. Future research should continue to prioritize human-in-the-loop evaluations, incorporating diverse perspectives and use cases to ensure the broad applicability and reliability of language models.

The integration of a weighted distribution of RAG with the Llama Large model represents a significant advancement in the pursuit of more accurate and reliable language models. The substantial improvements in performance metrics and the positive feedback from human evaluations demonstrate the potential of this approach to address the inherent limitations of traditional LLMs. By leveraging the strengths of both retrieval and generative paradigms, the weighted RAG mechanism paves the way for more sophisticated and contextually aware AI systems. The ongoing refinement and adaptation of this approach will likely drive further advancements in the field, fostering the development of more trustworthy and effective language models for a wide range of applications. The ongoing challenges and limitations identified in the error analysis highlight the importance of continuous improvement and innovation. Future research should focus on enhancing the context-awareness and adaptability of the weighting mechanism, exploring new methodologies for handling ambiguous queries, and integrating more sophisticated disambiguation strategies. By addressing these areas, researchers can further optimize the performance of weighted RAG models, ensuring their reliability and effectiveness in an ever-expanding array of applications. The pursuit of more accurate and contextually aware language models will undoubtedly continue to shape the future of NLP, driving progress and innovation in this dynamic and rapidly evolving field.

6. Conclusion

The research presented in this paper demonstrates significant advancements in enhancing the factual accuracy and contextual relevance of the Llama Large language model through the integration of a weighted distribution of Retrieval-Augmented Generation (RAG). The experimental results showed substantial improvements in precision, recall, F1 score, and BLEU score, confirming the efficacy of the weighted RAG mechanism in prioritizing high-quality information during the generation process. The positive feedback from human evaluations further validated the practical applicability and reliability of the enhanced model, highlighting its potential for deployment in various high-stakes environments where accuracy is crucial.

The contributions of this research extend beyond the immediate improvements in model performance. By introducing a sophisticated weighting mechanism for retrieved documents, the study provides a scalable framework that can be adapted to other language models and applications. This framework not only addresses the inherent limitations of traditional LLMs but also opens new avenues for research in dynamic context-aware weighting and real-time feedback integration. The findings underscore the importance of combining retrieval and generation paradigms to leverage their respective strengths, thereby advancing the state of the art in natural language processing.

Future work should focus on refining the weighting mechanism to enhance its adaptability and context-awareness. Exploring advanced retrieval algorithms and integrating semantic understanding could further mitigate the issues identified in the error analysis, such as the misinterpretation of ambiguous queries and over-reliance on high-weight documents. Additionally, expanding the application of the weighted RAG mechanism to multilingual settings and domain-specific corpora could significantly broaden its impact and utility. Addressing these open questions will be crucial for developing more robust and versatile language models, driving continued innovation and progress in the field of natural language processing.

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